

FREE • OPEN SOURCE

Python for H&H Modeling & Analysis

A hands-on course for water resources engineers. No prior programming experience required.

● Mohsen Tahmasebi Nasab, PhD

Swipe →

You don't need to become a **programmer.**

You just need Python to help solve your problems.

WHAT CHANGED

LLMs lowered the barrier to learning and using code.



Explain unfamiliar code



Debug errors faster



Turn plain descriptions into code

ChatGPT, Claude, Gemini, and Copilot are AI assistants powered by Large Language Models (LLMs).

THE APPROACH

Built for engineers.

Real H&H workflows

Every example is hydrology and hydraulics, not toy exercises.

Clarity over cleverness

Readable code a first-time user can actually follow.

LLM assistance from day one

Focus on solving engineering problems, not memorizing syntax.

SIX MODULES

From basics to real workflows.

01

Setup & Fundamentals

Python via H&H
analogies

02

Time Series

Hydrographs, flow stats

03

Vector Data

Watersheds, mapping

04

Raster Data

DEMs, slope, land cover

05

USGS & Stream Data

Gauge data retrieval

06

Precipitation

NOAA Atlas 14 storms

BY THE END

You'll automate the tedious work.



Pull USGS gauge data in minutes



Generate design storms from NOAA



Map watersheds, terrain & land cover



Process time series and Excel data programmatically

HOW YOU LEARN

Paste the code.
Ask why.
Keep going.

Every module has ready-to-use prompts
and LLM-assisted debugging exercises built in.

ChatGPT • Claude • Gemini • Copilot



ONE IMPORTANT CAVEAT

An LLM assistant is a study buddy, not the engineer.



Great for learning, explaining, and debugging



You own the assumptions, methods, and QA/QC



Always sanity-check what it gives you

LLM-assisted coding can help you learn faster. The engineering judgment stays yours.

START TODAY

Free. Open. No installation required.

Runs entirely in Google Colab

Beginner friendly

Pilot-tested with WRE professionals

Find it on GitHub:

github.com/mohsennasab/python-fundamentals-hh

● hydromohsen.com

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